Developing daily gap-filled chlorophyll-a datasets using deep neural networks with co-located environmental variables

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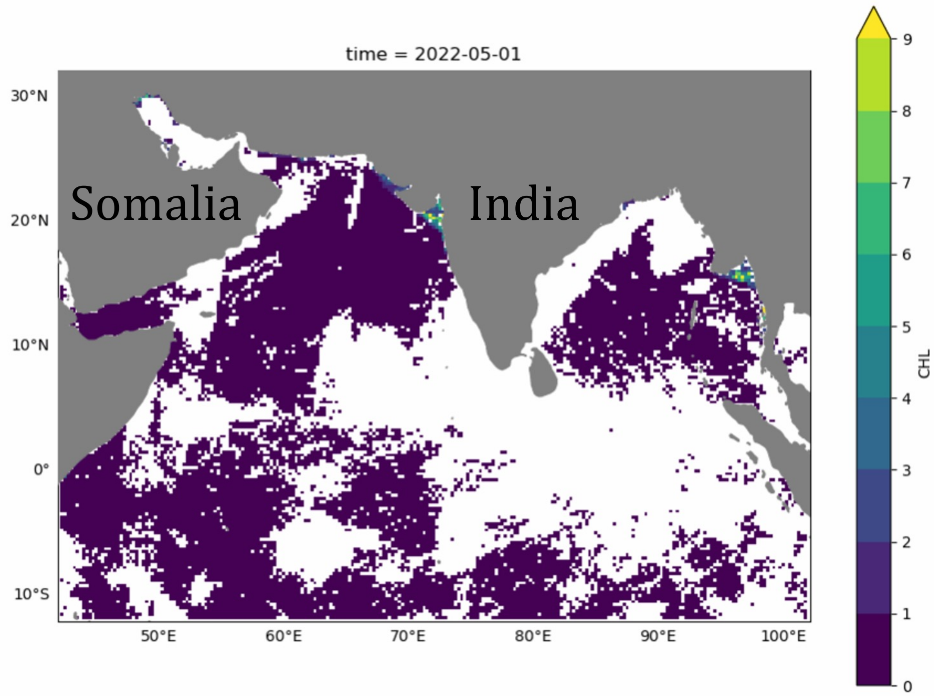
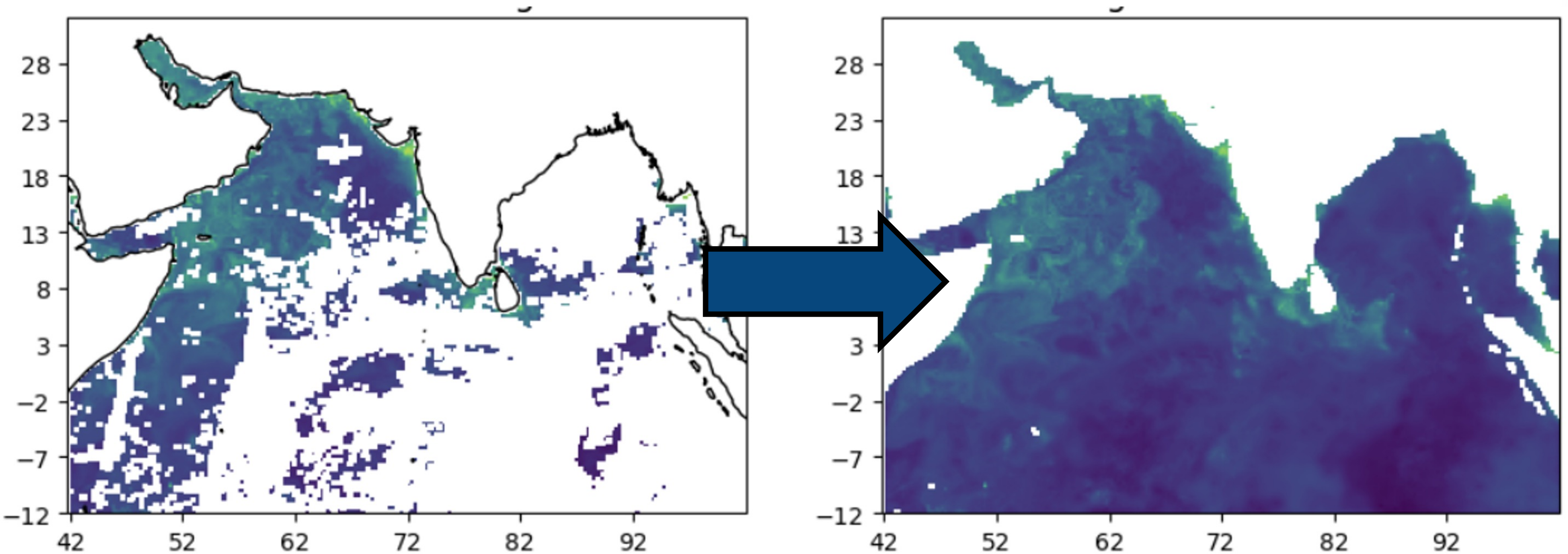
The Problem: Missing Chlorophyll Data

The Arabian Sea and Bay of Bengal exemplify the problem: white are clouds (NAs)

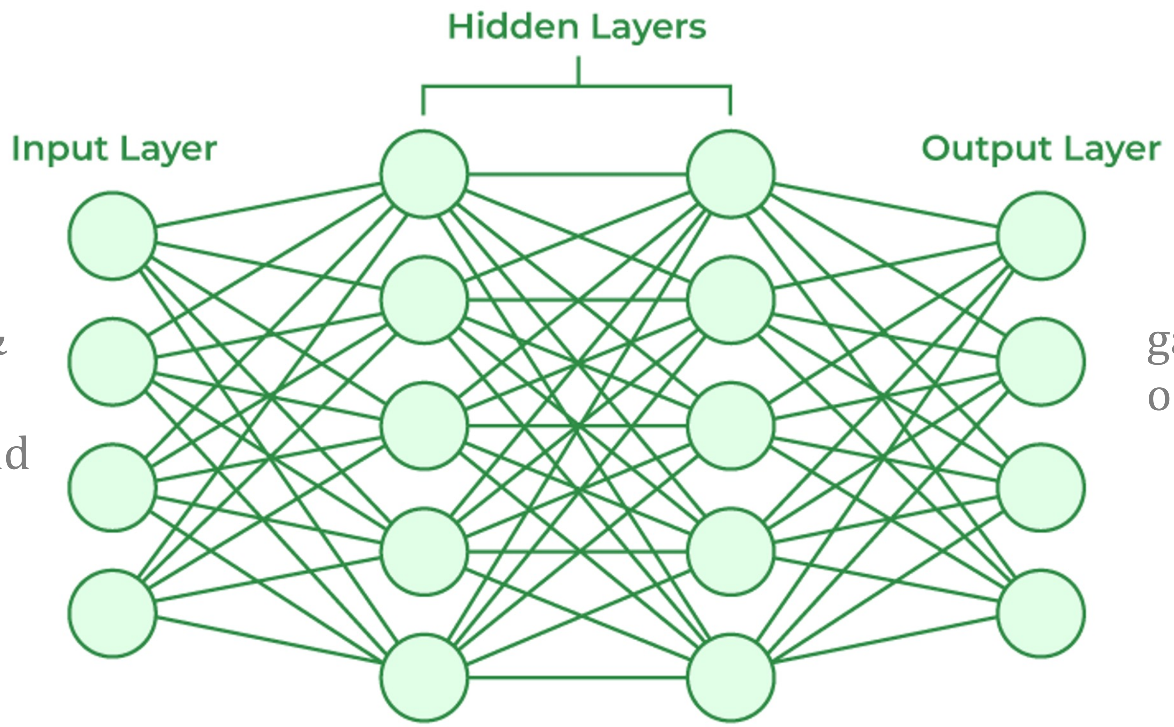


Solution: Gap-filling algorithms



Our work: comparing two different deep neural nets for gap-filling

Observed Chl-a & Environmental variables on a grid



gap-filled Chl-a on a grid

Why use neural nets?

1. Universal Function Approximation Theorem

Any non-linear relationship can be arbitrarily approximated by a suitable neural network of ONLY one hidden layer.

2. Advantages of Deep Networks

By extending past one hidden layer, we can learn hierarchical features automatically, and scalably.

3. Convolutional Neural Networks (CNNs) for Images

Learn local patterns and features in the image. Better fit for image data compared to standard algorithms that are used in traditional regression techniques.

Adding Environmental Variables to neural networks

This is a new area of research for neural networks with earth sensing data.

1. Environmental variables affect Chlorophyll-a concentrations
2. Neural networks: able to learn non-linear and spatial relationships
3. Maximize the use of all information for better predictions
4. May provide a new approach for making Chl-a predictions when there is no Chl-a observations (pre-1997)

What variables to use?

Random Forest and Gradient-Boosted Trees were used to explore the important variables:

- SST: Sea Surface Temperature
- N-S Wind (uwind): North-South Wind (Eastward Wind)
- E-W Wind (vwind): East-West Wind (Northward Wind)
- adt: Absolute Dynamic Topography (sea surface height)
- air_temp: Air Temperature
- curr_dir: Current Direction
- curr_speed: Current Speed
- mlotst: Mixed Layer Depth
- sla: Sea Level Anomaly
- so: Salinity

From a combination of sources. [Link to Indian Ocean dataset documentation]

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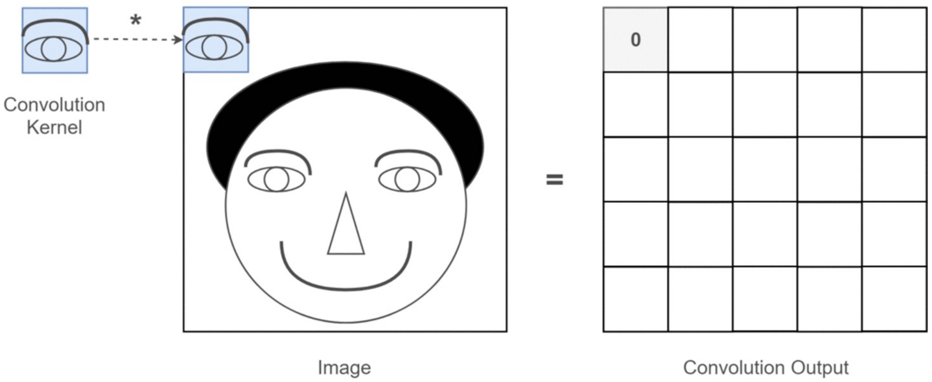
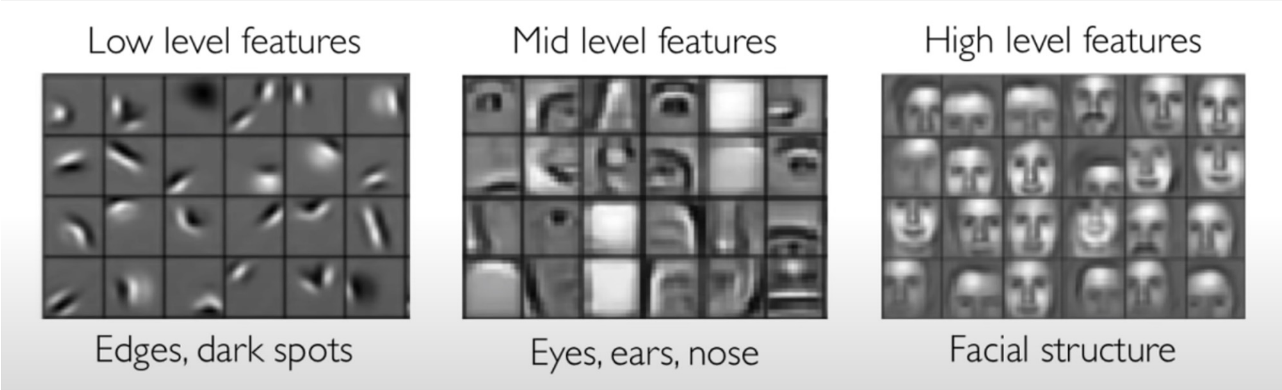
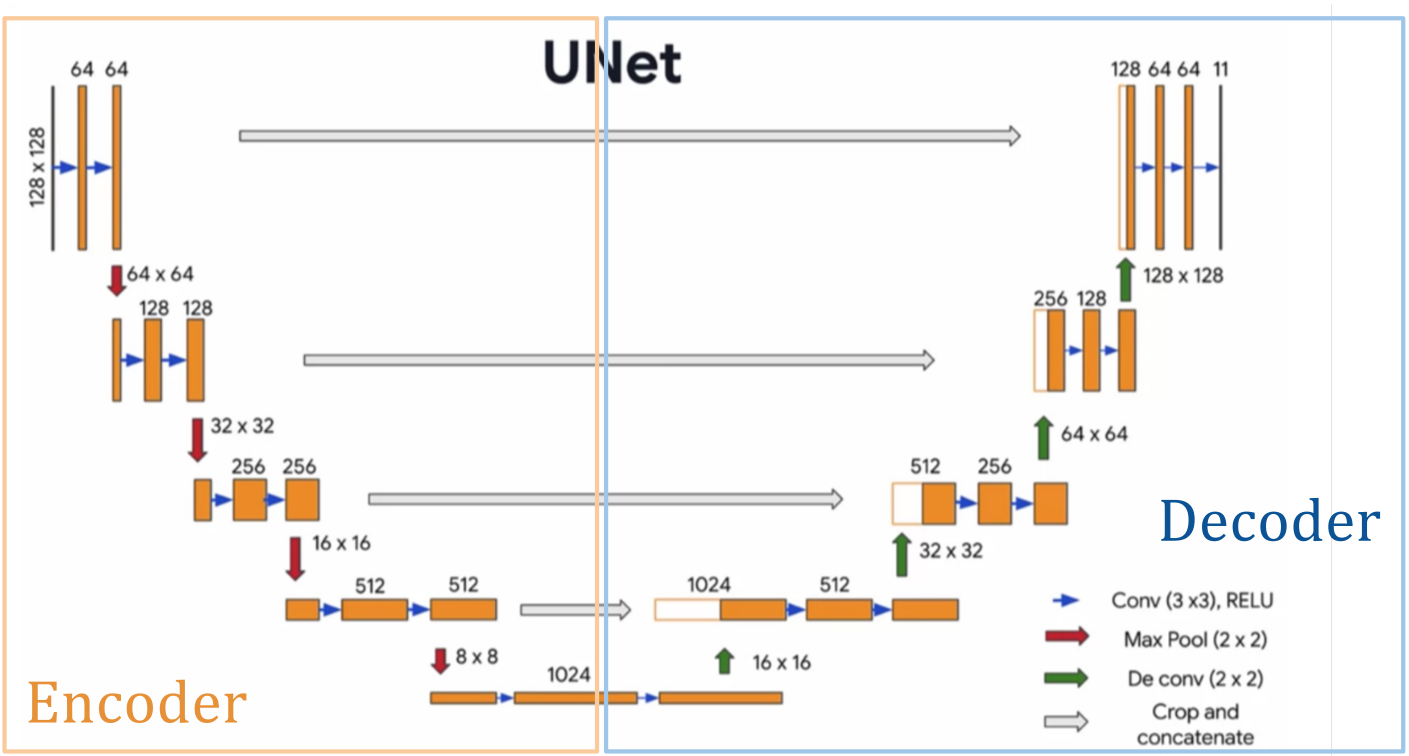
CNN models: a type of neural network

- Local Feature Detection
- Features are LEARNED
- Hierarchical Learning





U-Net architecture: a type of CNN model

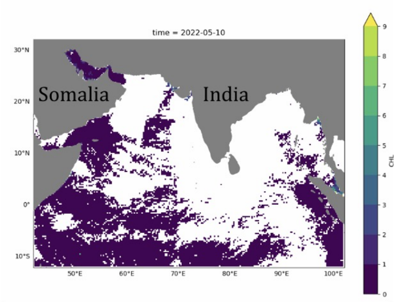


Training and testing the predictive model

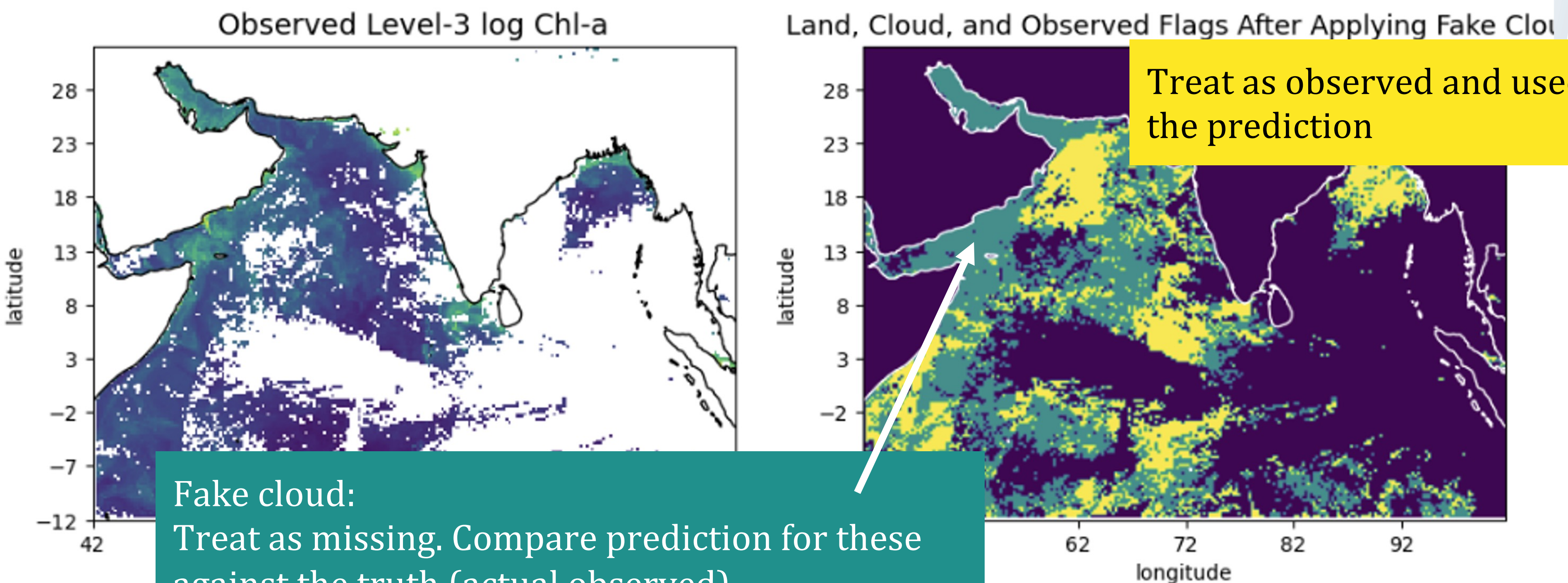
Data: Global Ocean Colour (Copernicus-GlobColour) level 3 with gaps

- Train (parameters)
- Validate (tune model structure)
- Test (compare predictions to held out data)

Trained on years 2015-2017
Validated on year 2018
Tested on year 2020



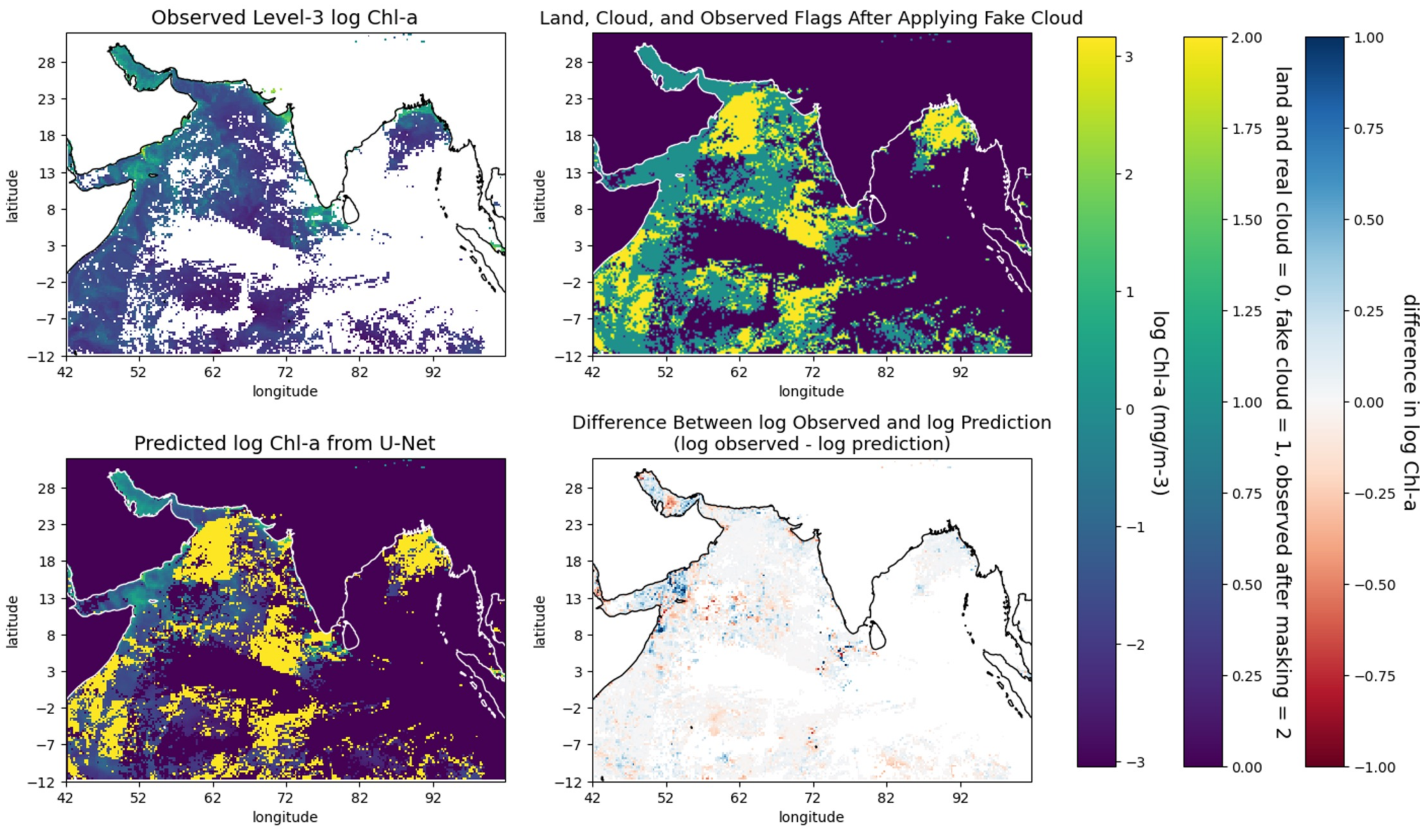
Testing: Using fake clouds to create a test set



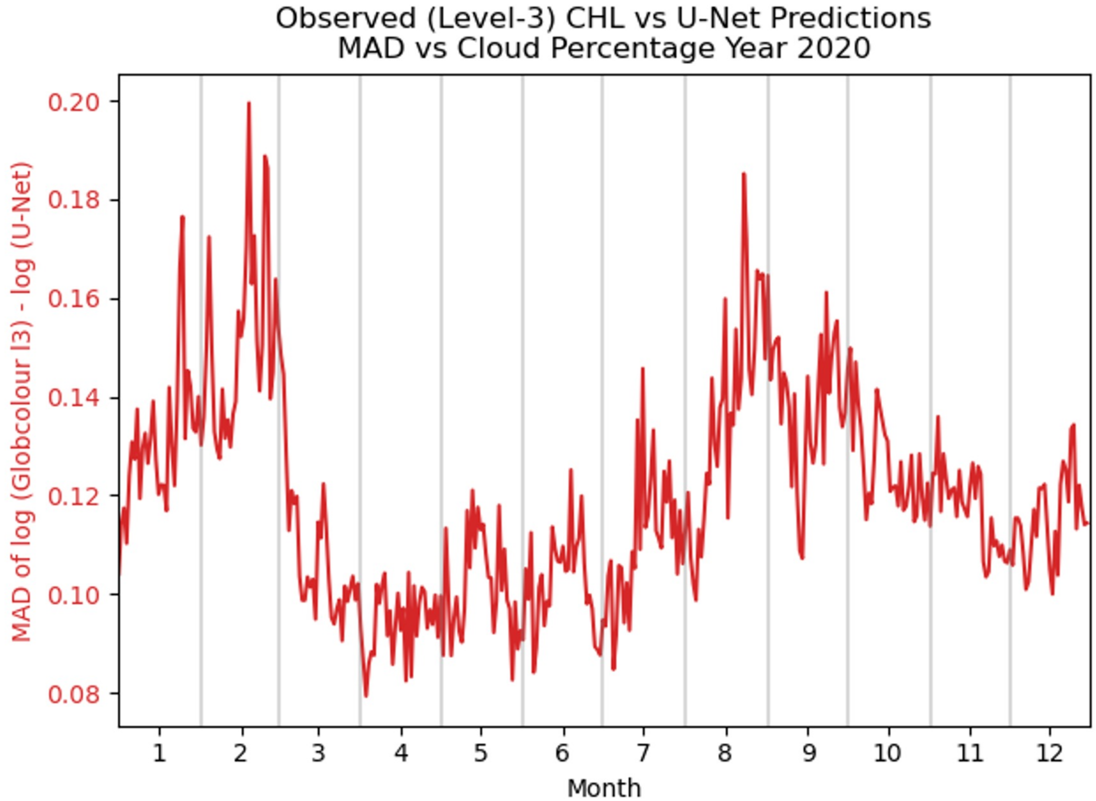
Treat as observed and used in the prediction

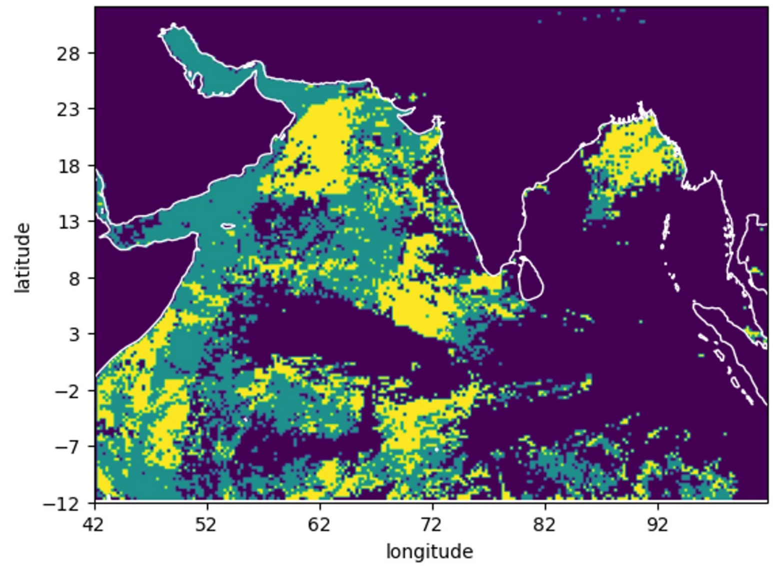
Fake cloud:
Treat as missing. Compare prediction for these against the truth (actual observed)

Test: comparison of predictions to held out data



Test: prediction versus true values over a year





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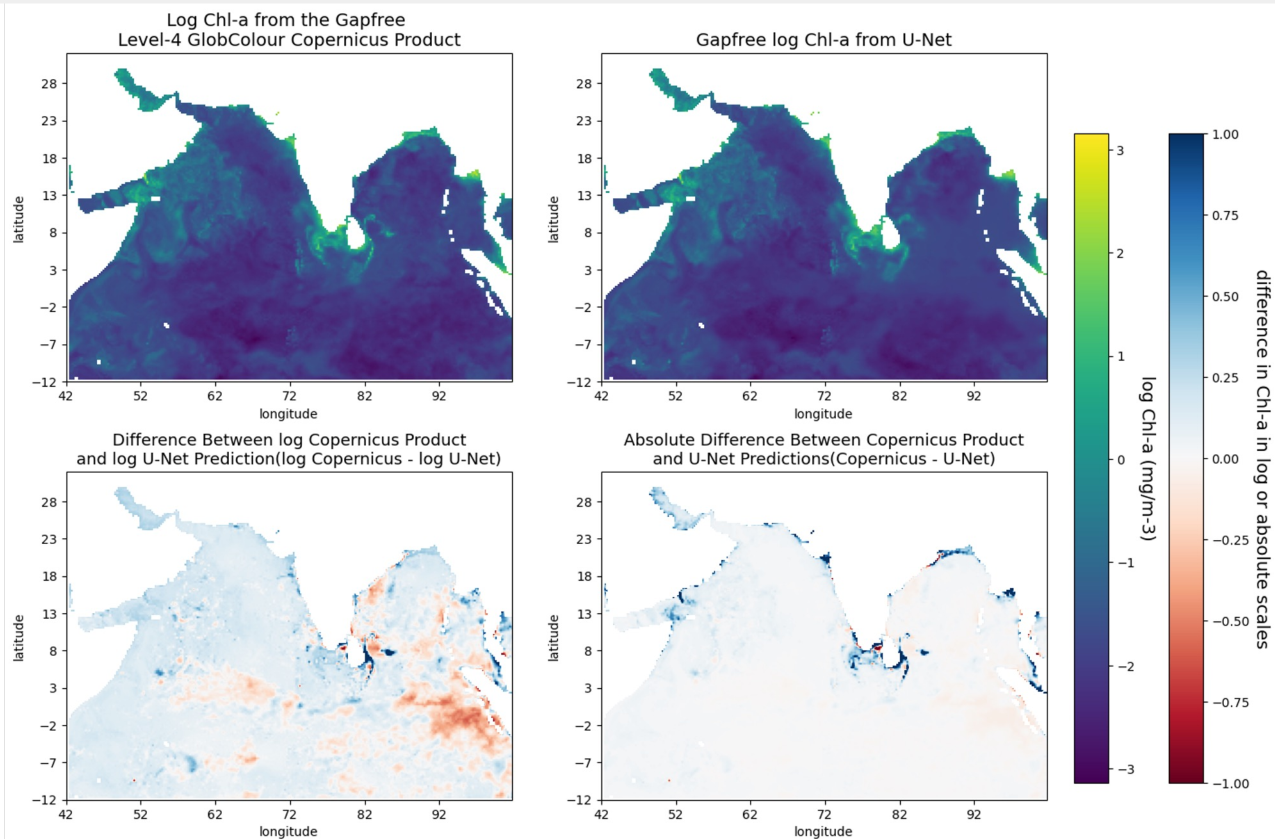
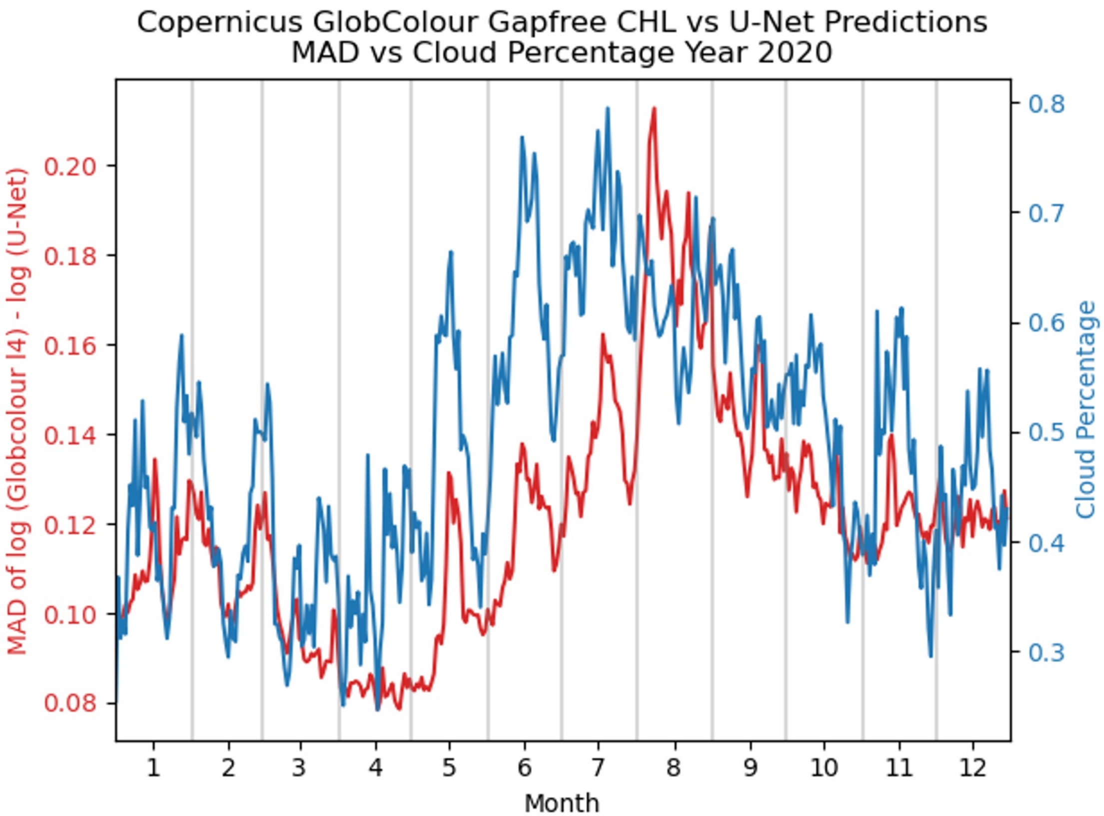
Comparison to Copernicus gapfree level 4

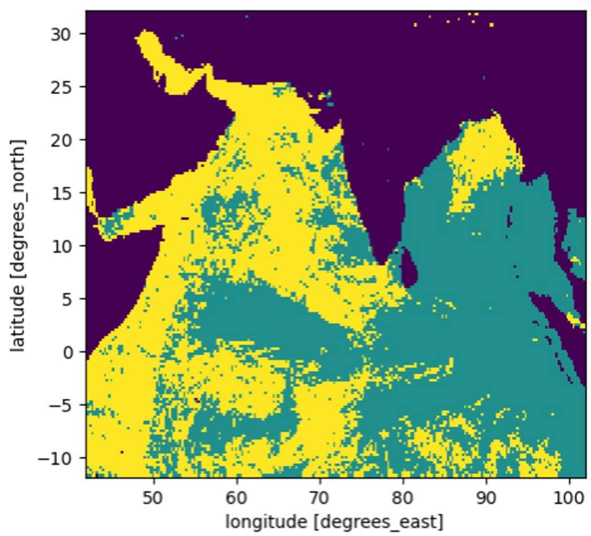


Global Ocean Colour (Copernicus-GlobColour) level 4 gapfree

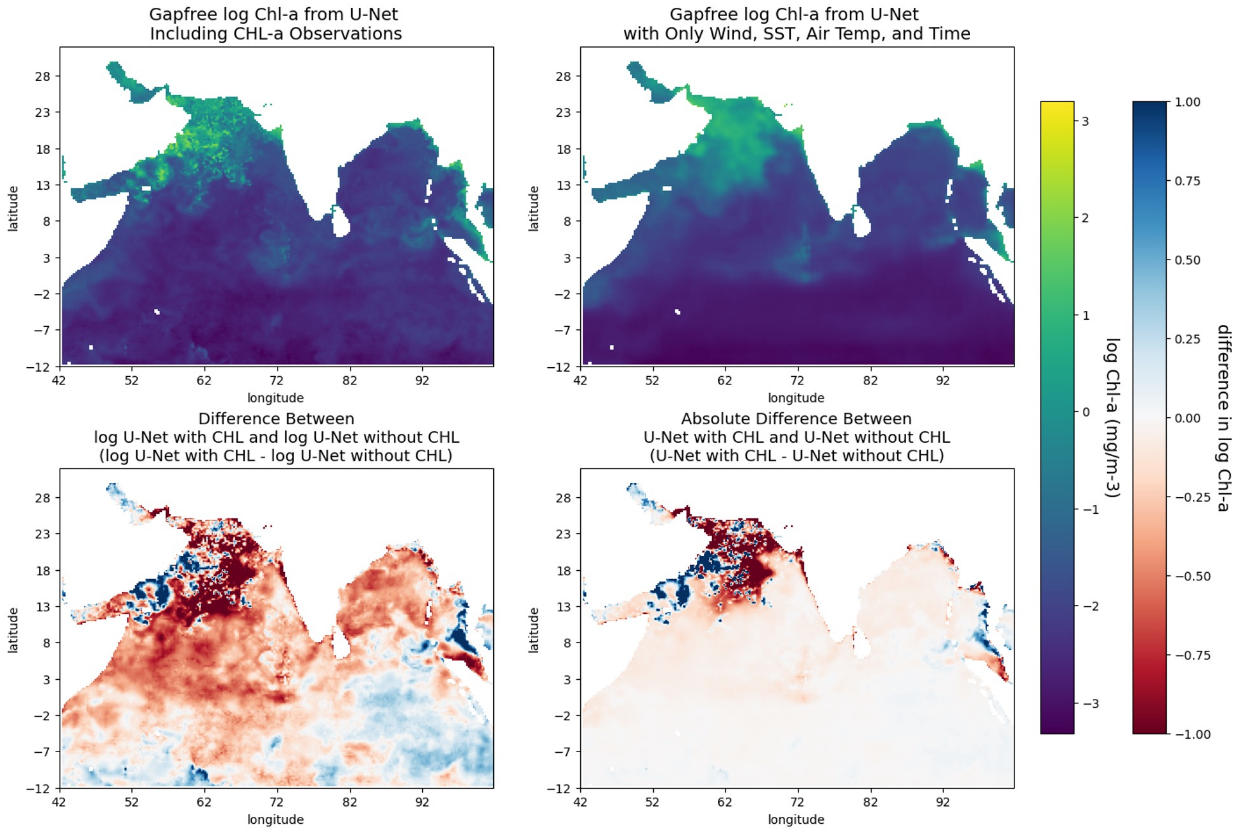
Main difference along coasts

Comparison to Copernicus gapfree over a year





Example results without Chl-a information



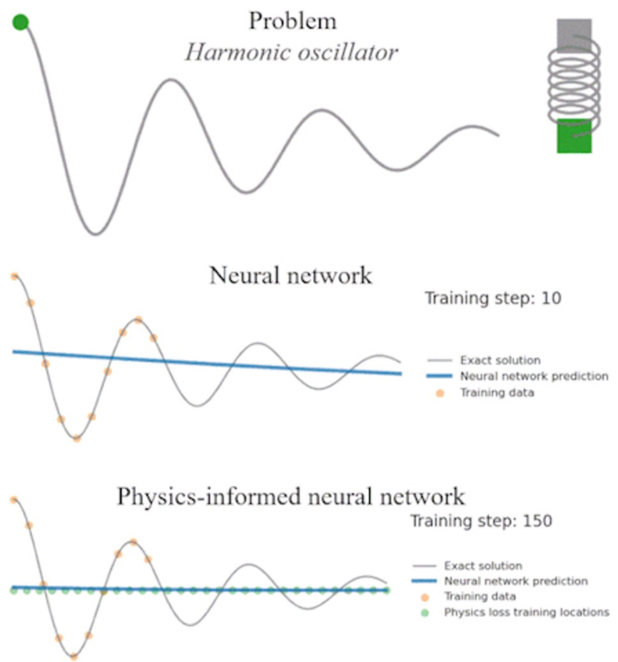
Work in progress!

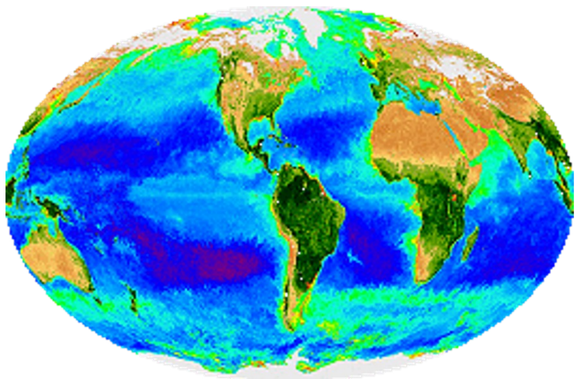
Although there is bias the wind, SST, air temperature, and season ALONE are able to capture the overall pattern.

U-Net Main Takeaways

- U-Net with Chl:
  - Works well (similar to Copernicus gapfree L4 globColour)
  - Mainly different along the coast
- U-Net without Chl:
  - Captures general patterns with only wind, SST, air temp and season

PINN models

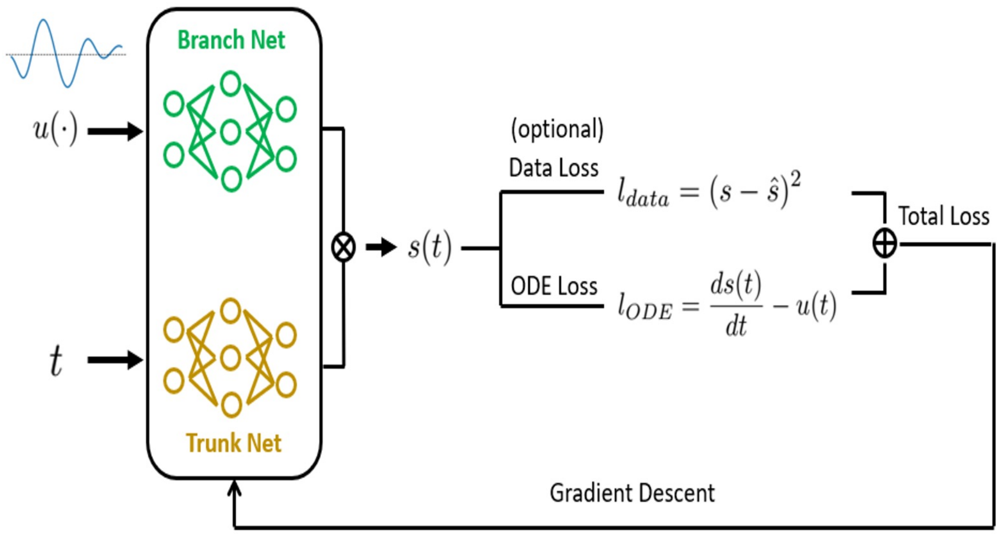




- Chlorophyll obeys physical laws!
- Data driven approaches do not!

PINN models

DeepONet (*Lu Lu, et al. 2019*) Chl PDE(*Matthias Onabid, 2017*)



+

$$\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} - \rho(x, y, z)$$

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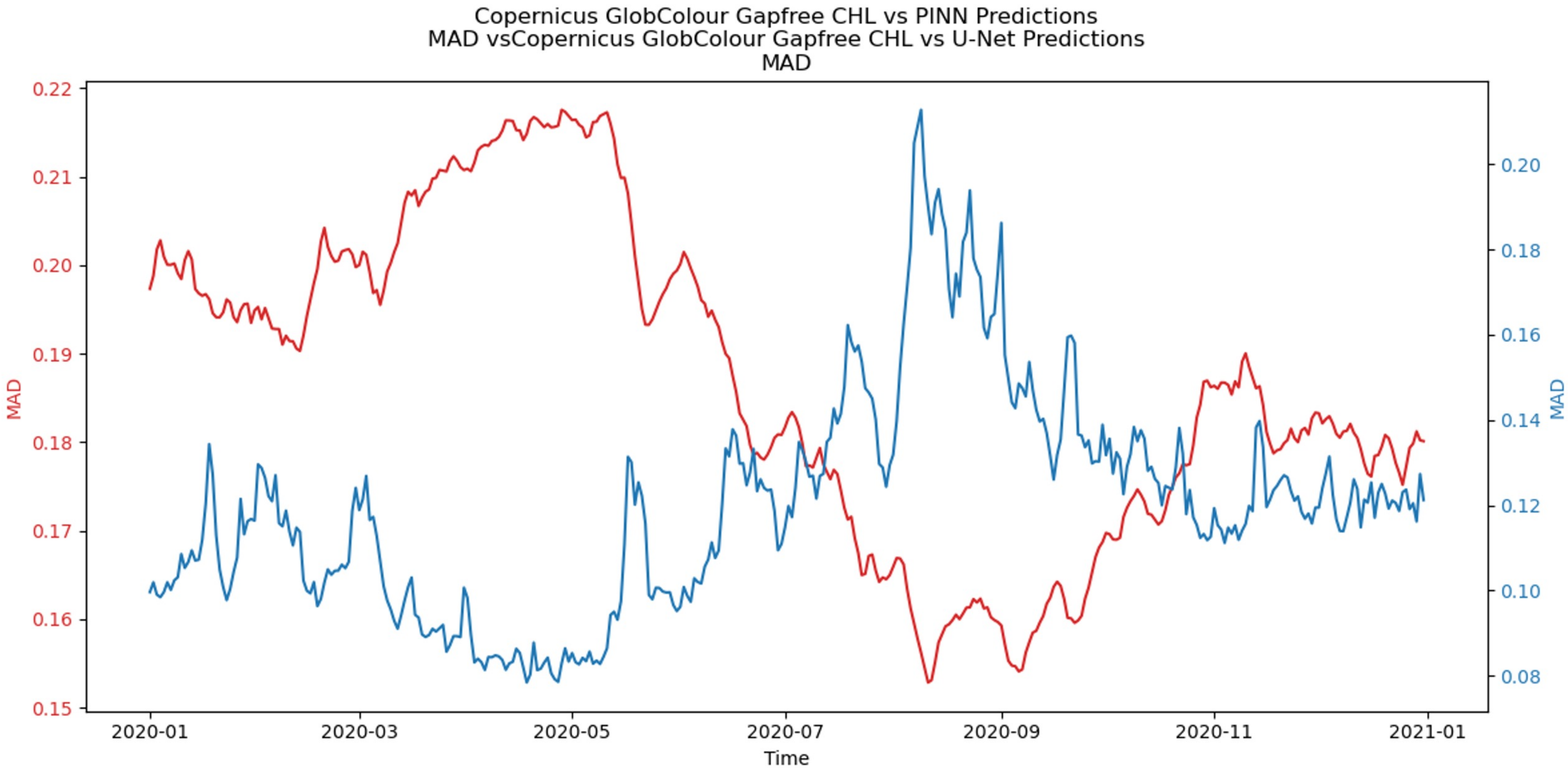
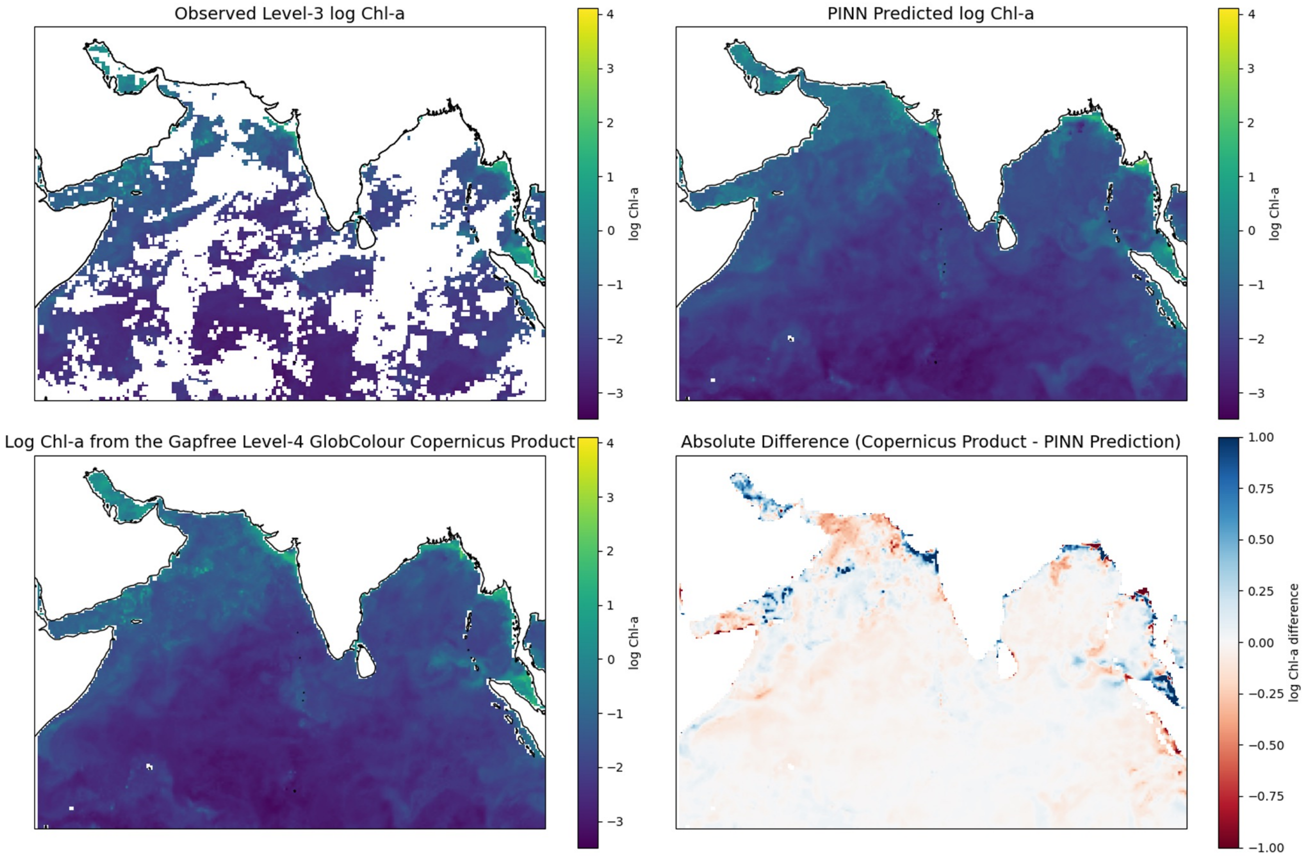
PINN: Results



PINN: Results



PINN Main Takeaways

- Performance:
  - Spatially similar to to Copernicus gapfree L4 globColour(Promising!)
  - Deviates from U-Net and Copernicus significantly in terms of overall value(Interesting!)